

Introduction to Earth System

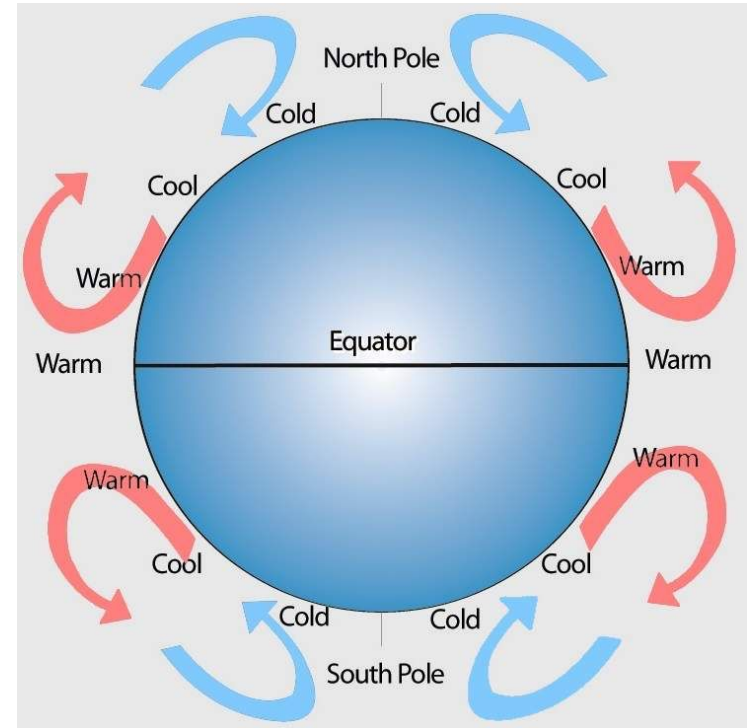
Dynamics of the atmosphere

Redistribution of the energy

Transfer of energy: from low to high latitudes via atmospheric and oceanic circulations

Energy redistribution occurs through:

- Conduction
- Convection
- Advection
- Latent heat

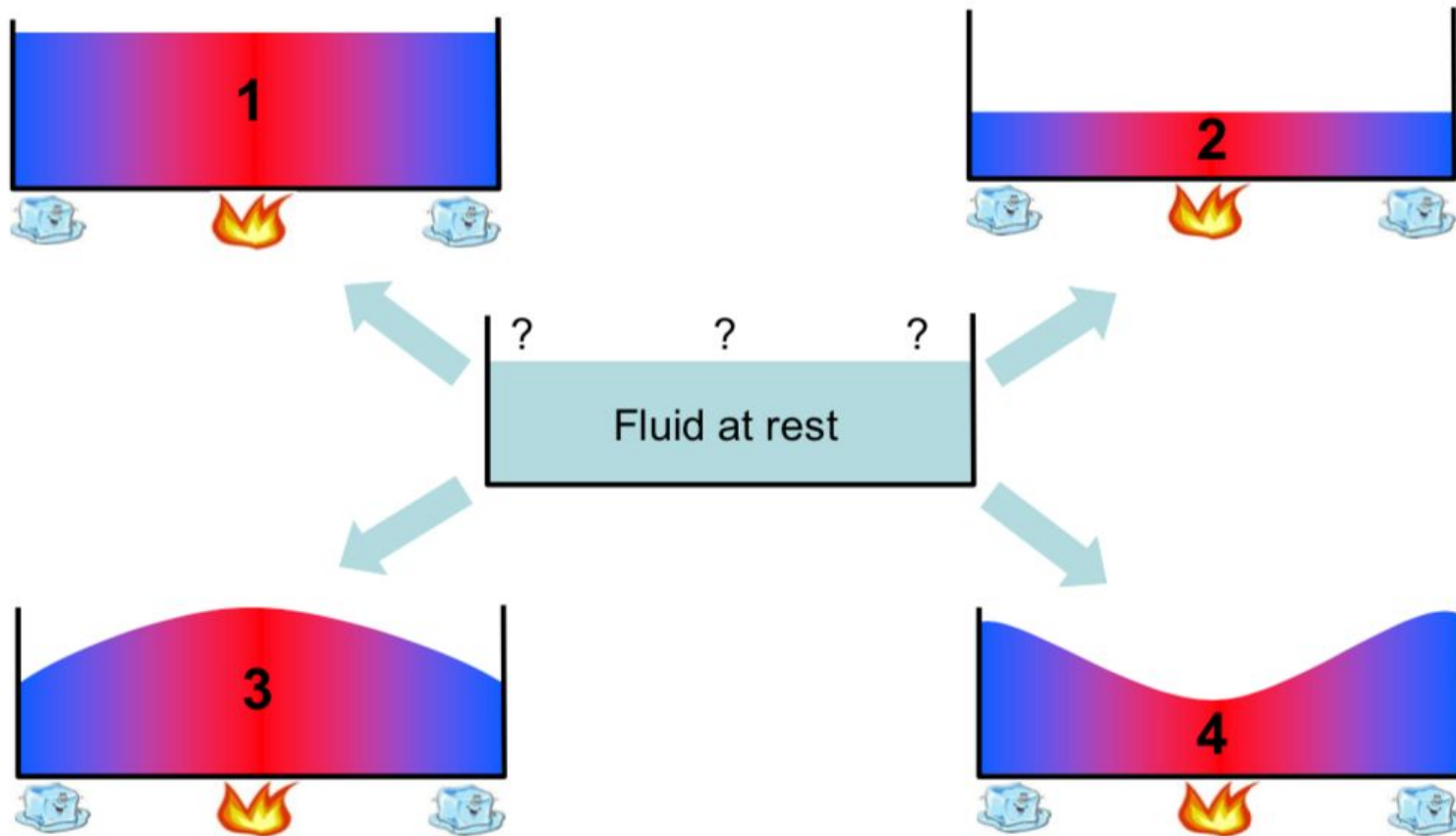


Observe the movements and global structures of the atmosphere

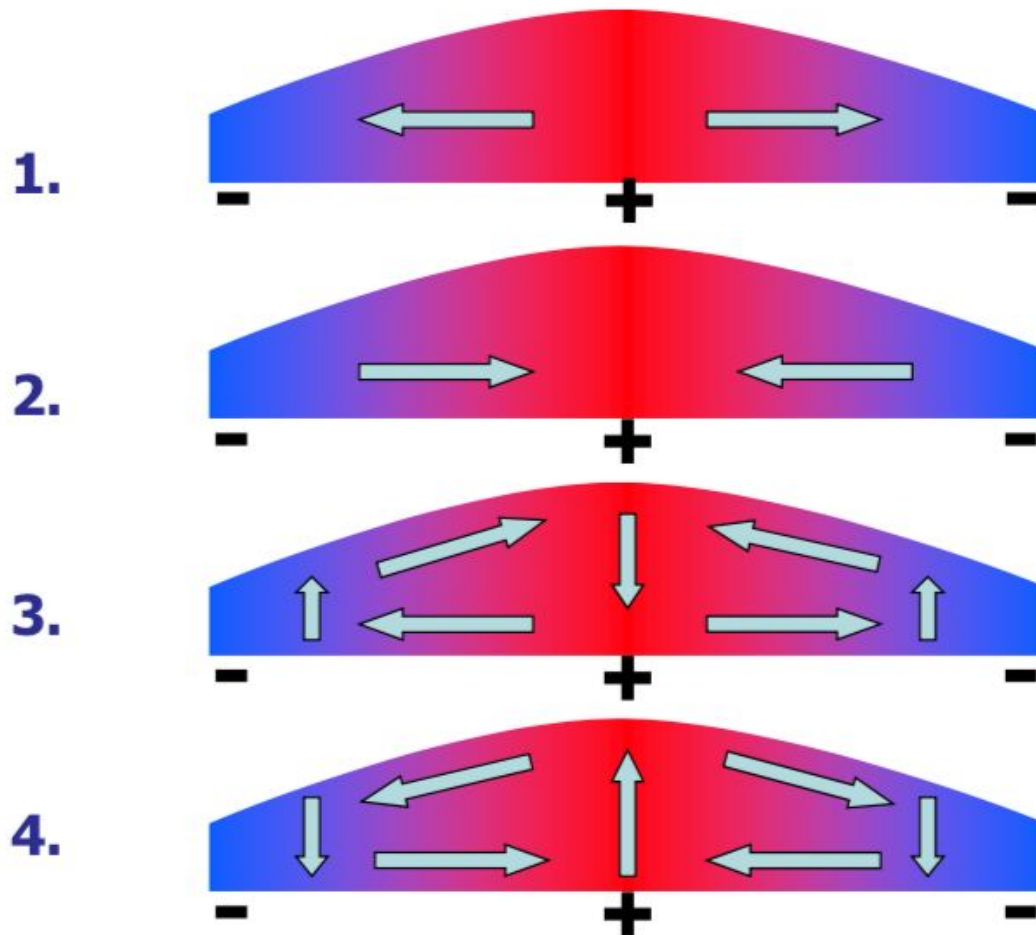


Example: the container has the two ends constantly cooled and the center heated

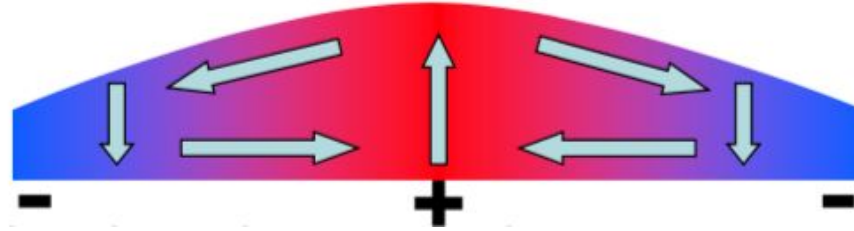
Q: What change can be observed?



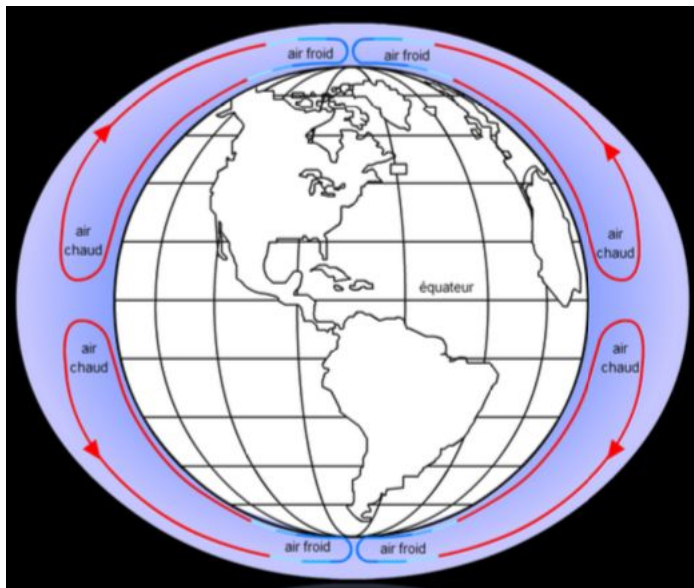
Q: Which sketch best illustrates the movements within the fluid?



Hadley's cell



- **Georges HADLEY** in 1735: a simple model of atmospheric circulation → Hadley's cell.
- Troposphere

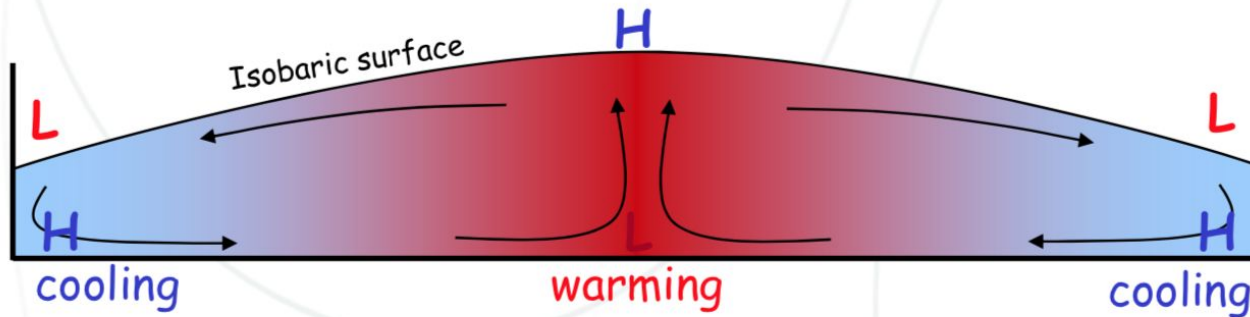


The Hadley cell works only for a non-rotating earth

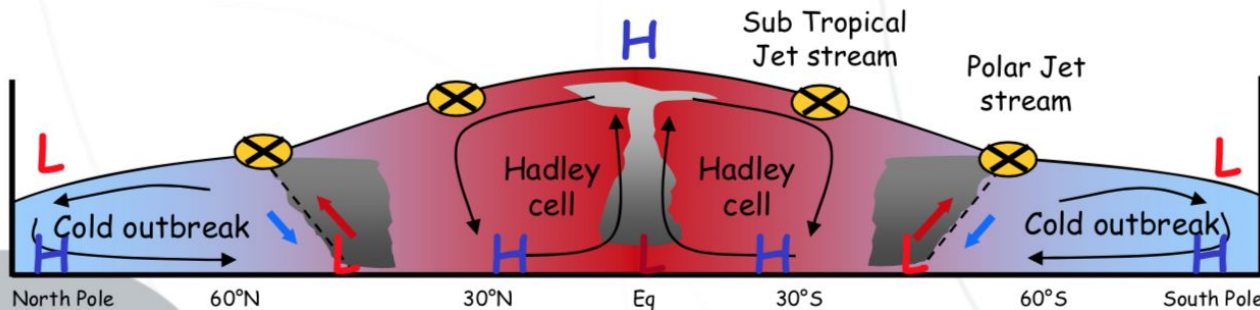
Origin of the General Circulation



Without energy exchange



With energy exchanges

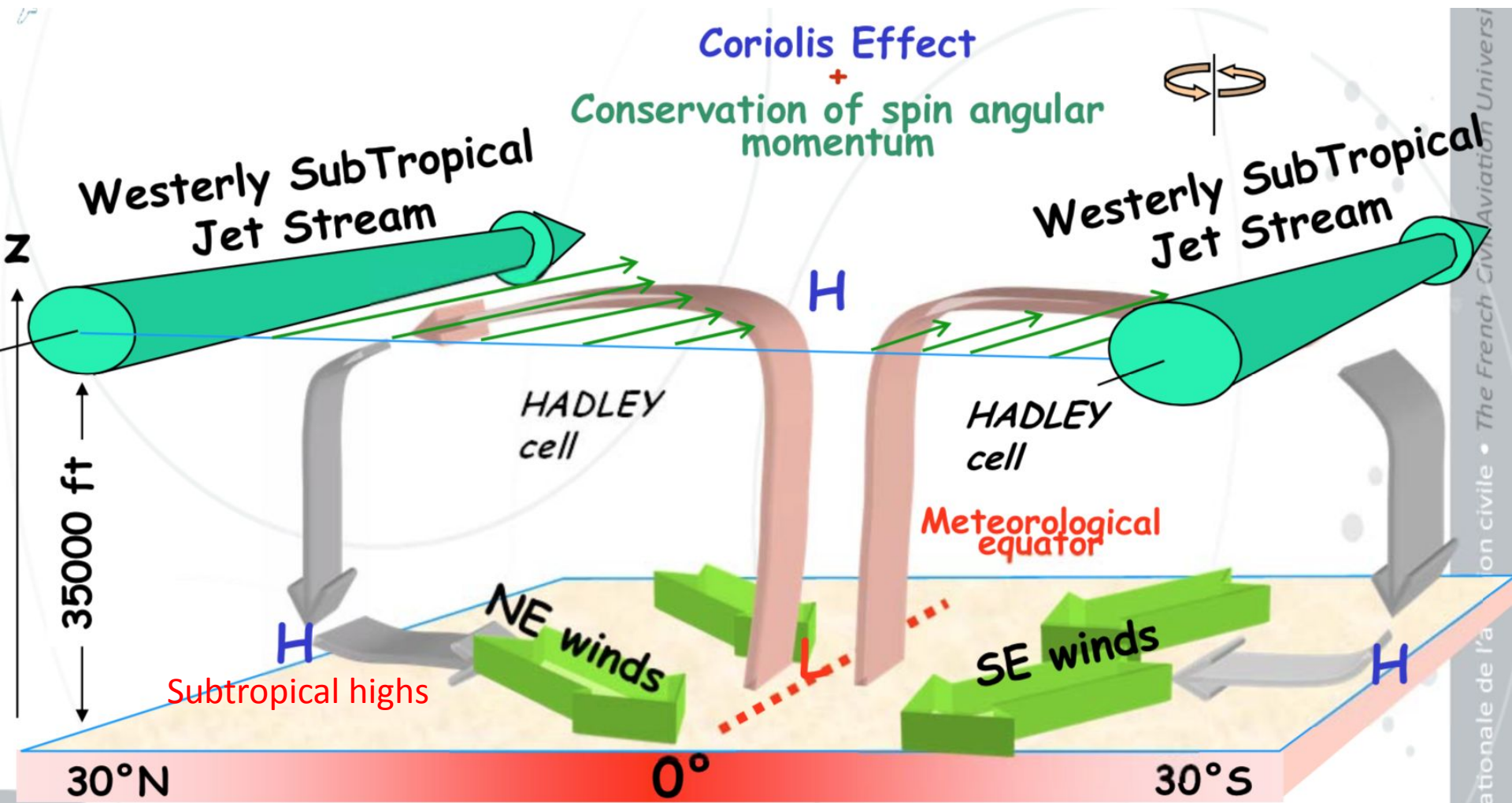


The Earth is round and turns on itself

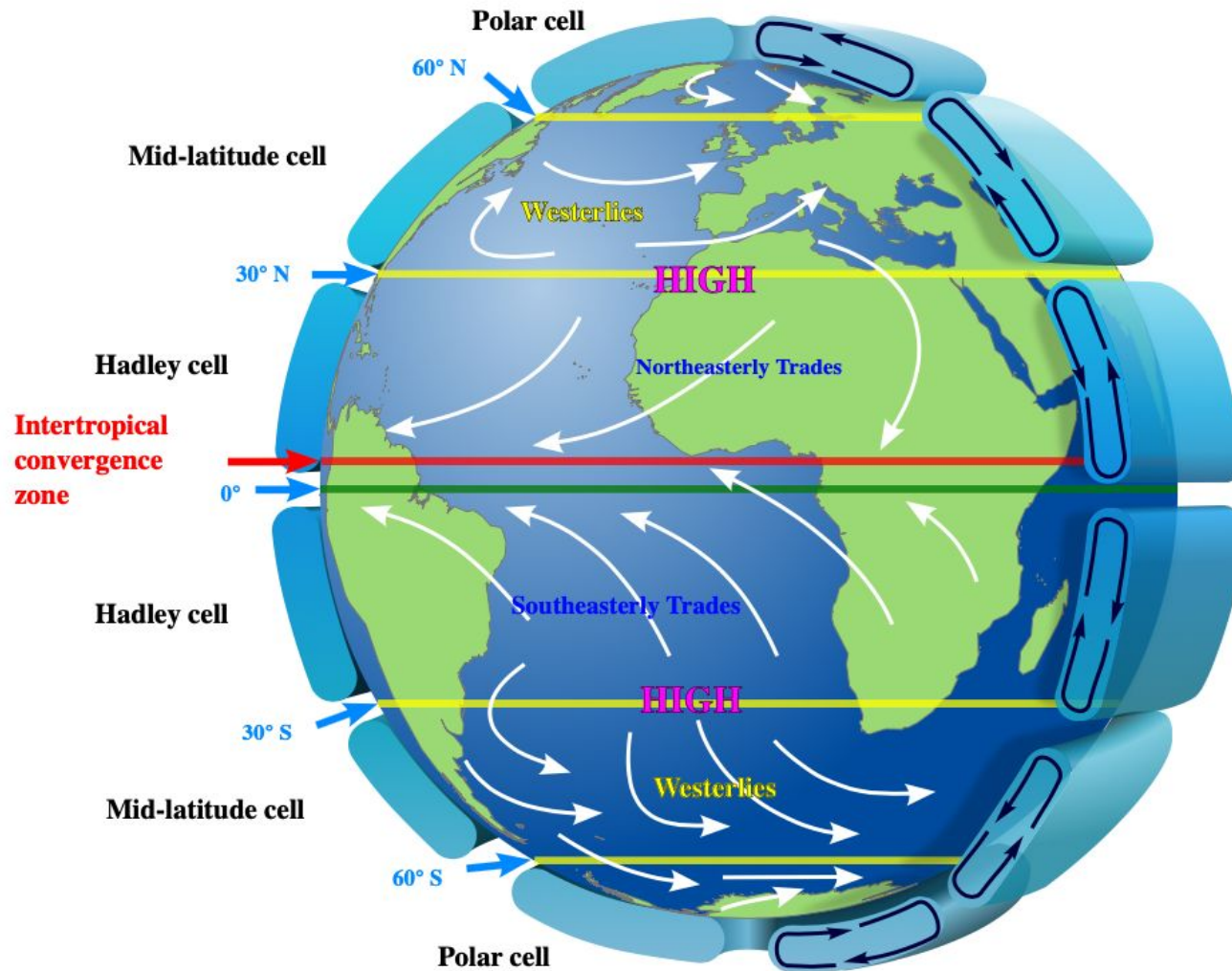
Q: Why is the Hadley's descending branch at about 30°N

Q: Why does sub-tropical jet stream exist?

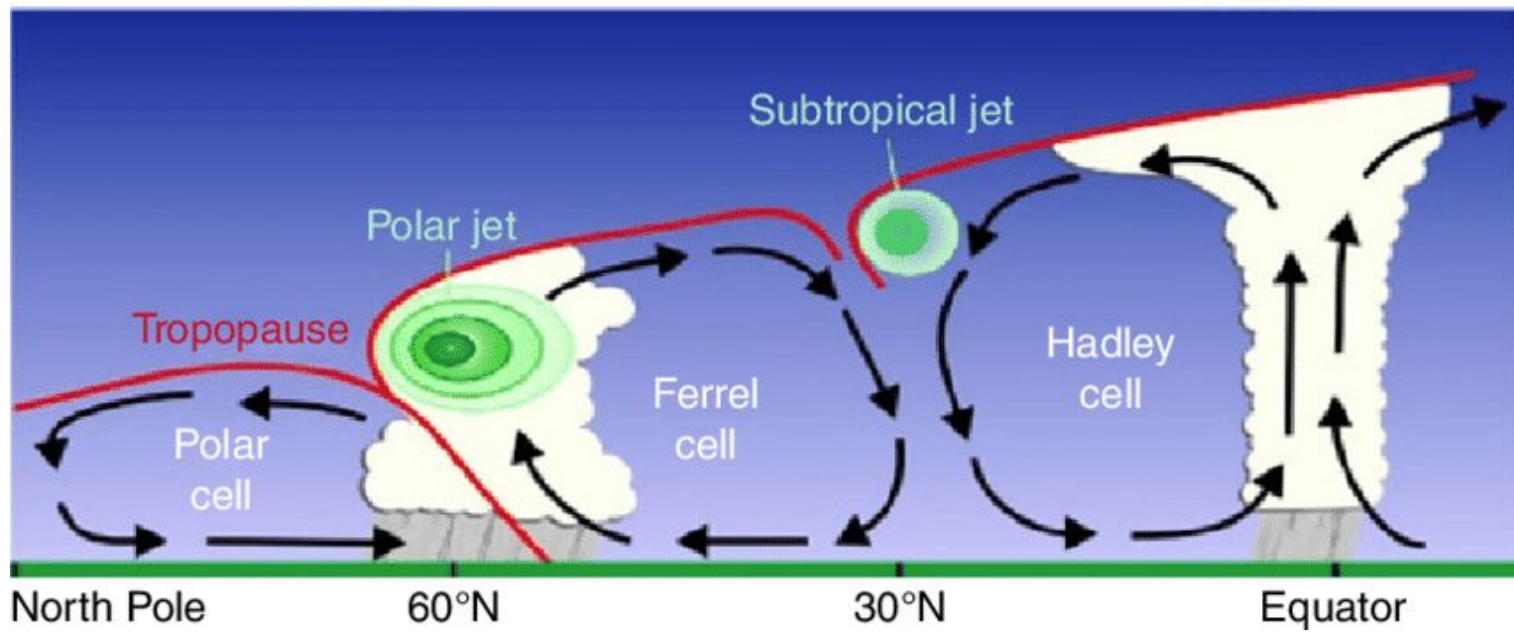
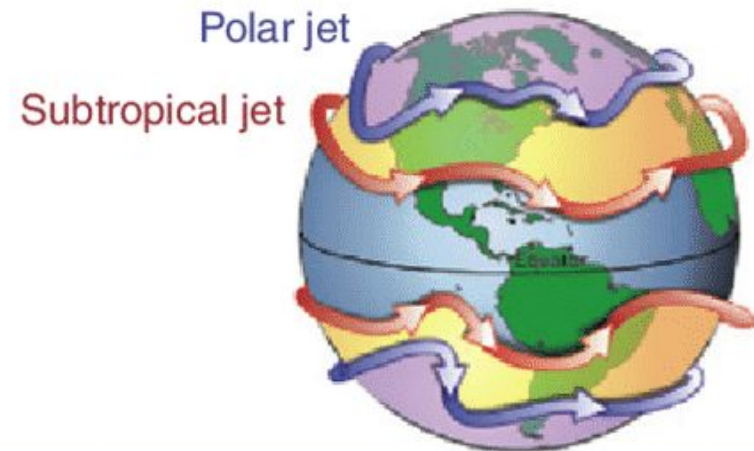
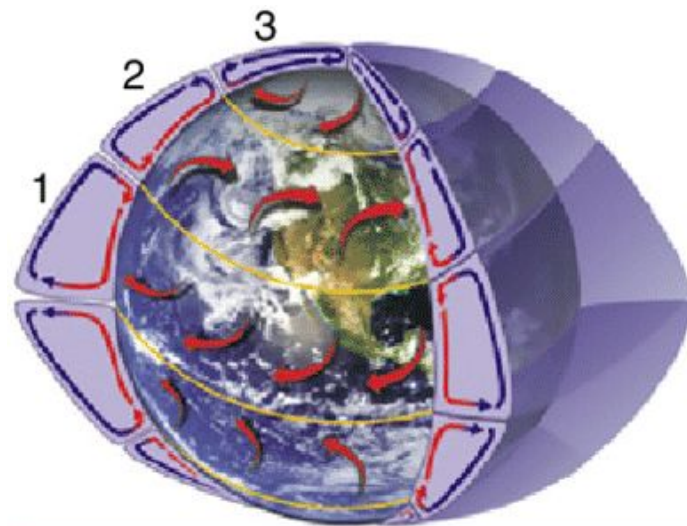
Zoom in tropical region (30°N-30°S)



JET = quasi-horizontal, flattened tubular air-current generally located near the tropopause, very unstable and characterized by high speeds (> 60 kt) and by strong vertical and horizontal shearing of the wind



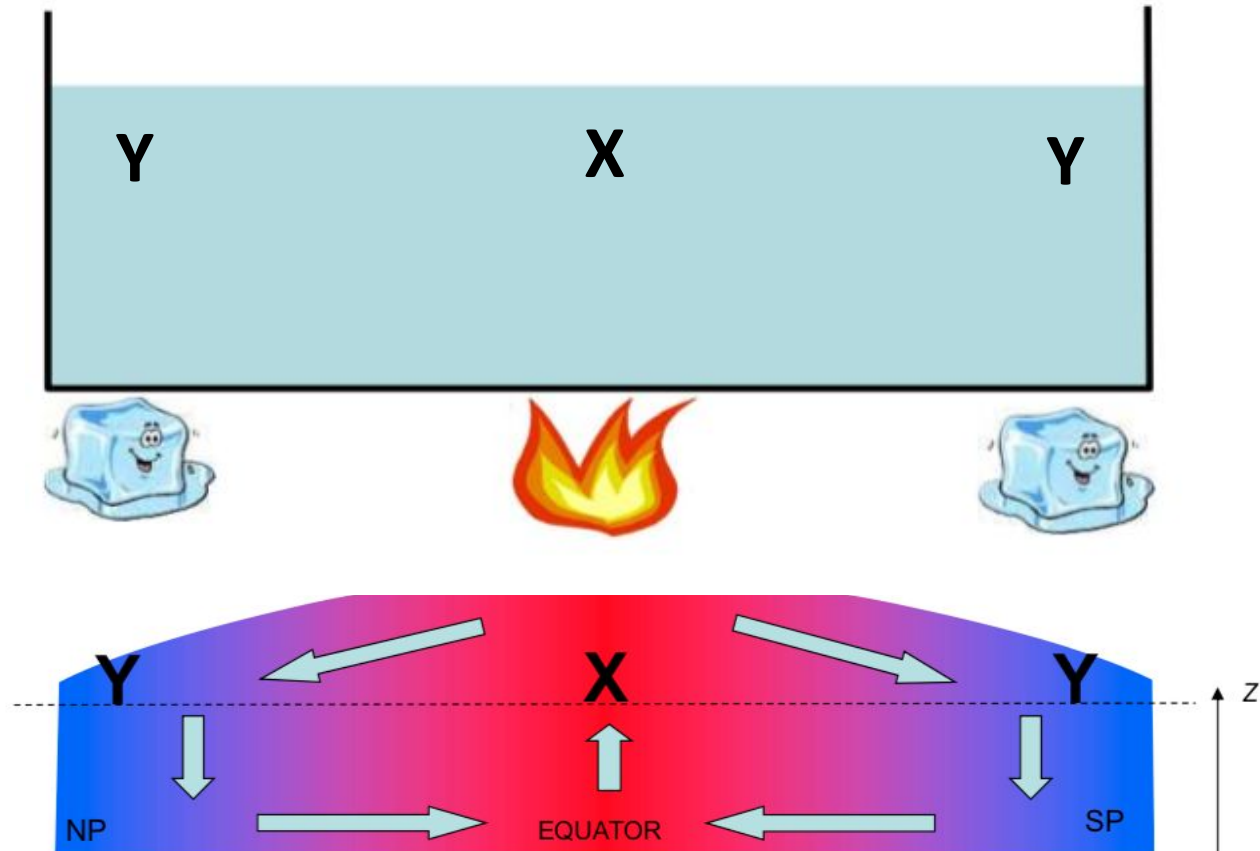
Global circulation of Earth's atmosphere displaying Hadley cell, Ferrell cell and polar cell.

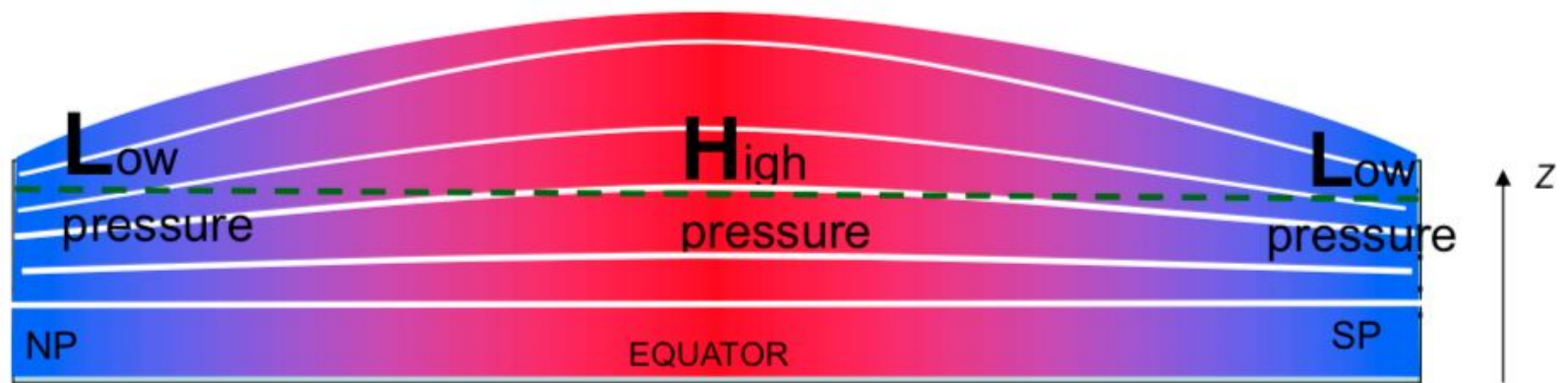
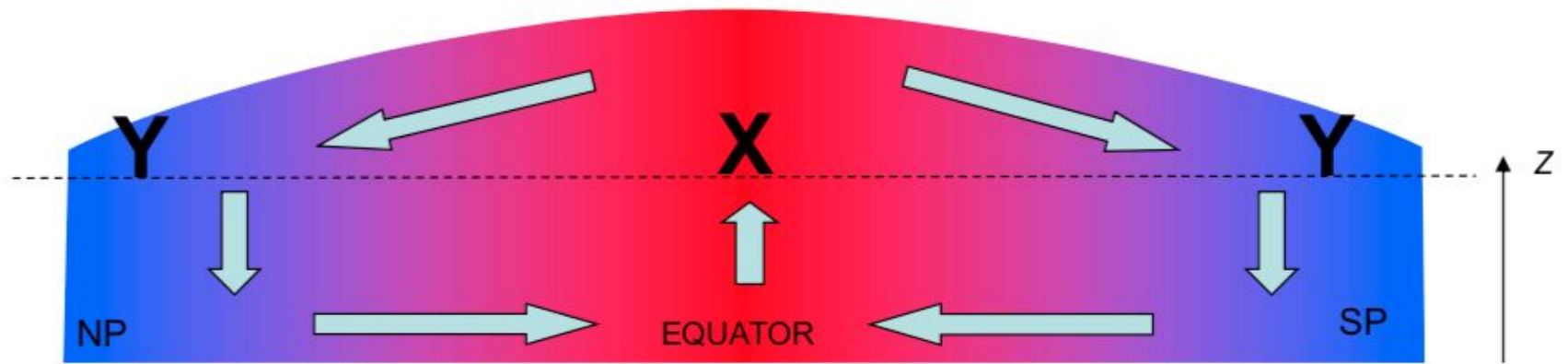


Description at upper level

Q: What can you say about the pressure field observed in X and Y, 2 points located at the same altitude in the troposphere?

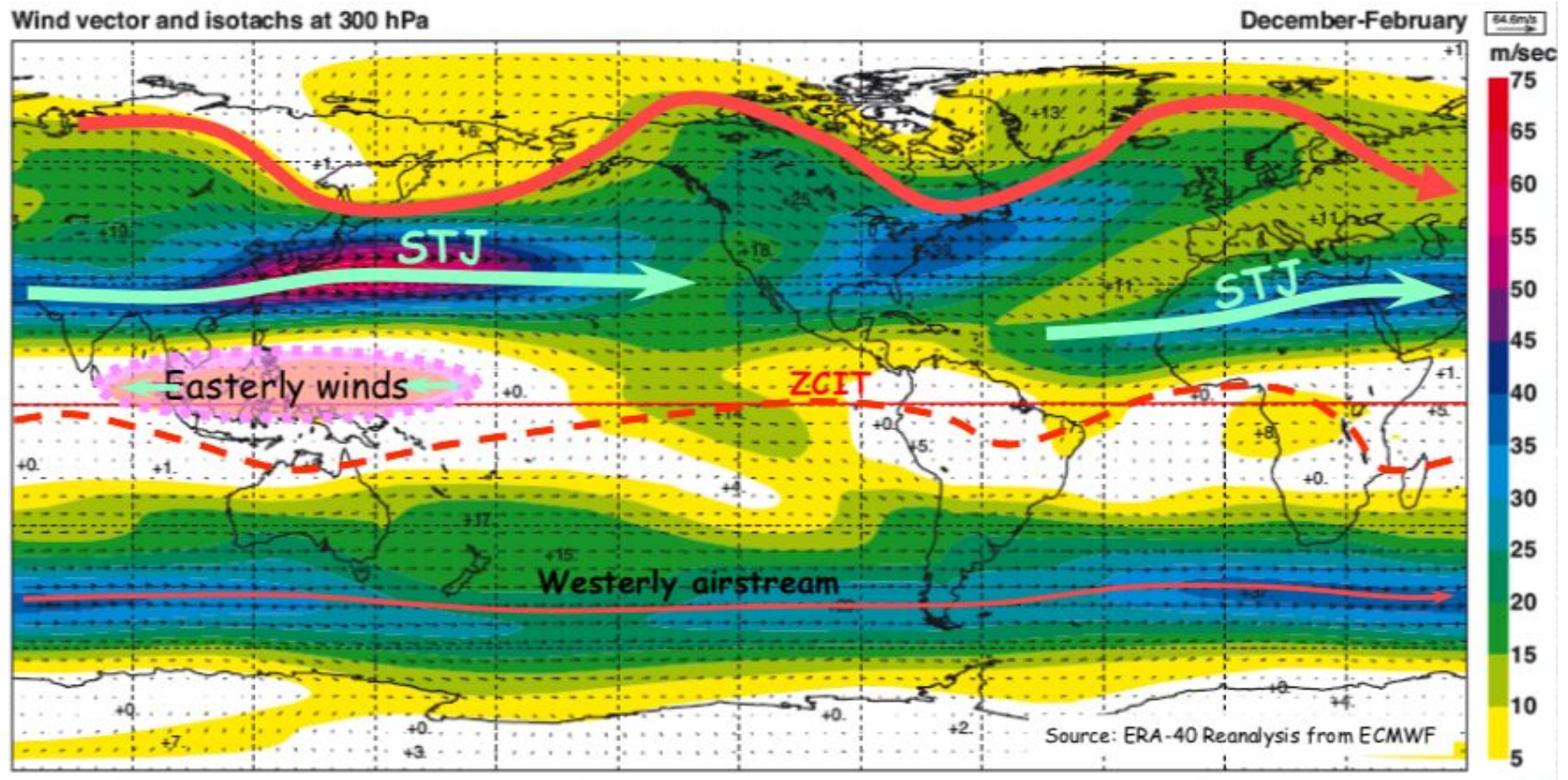
1. $P_x = P_y$
2. $P_x > P_y$
3. $P_x < P_y$





At upper level from December to February

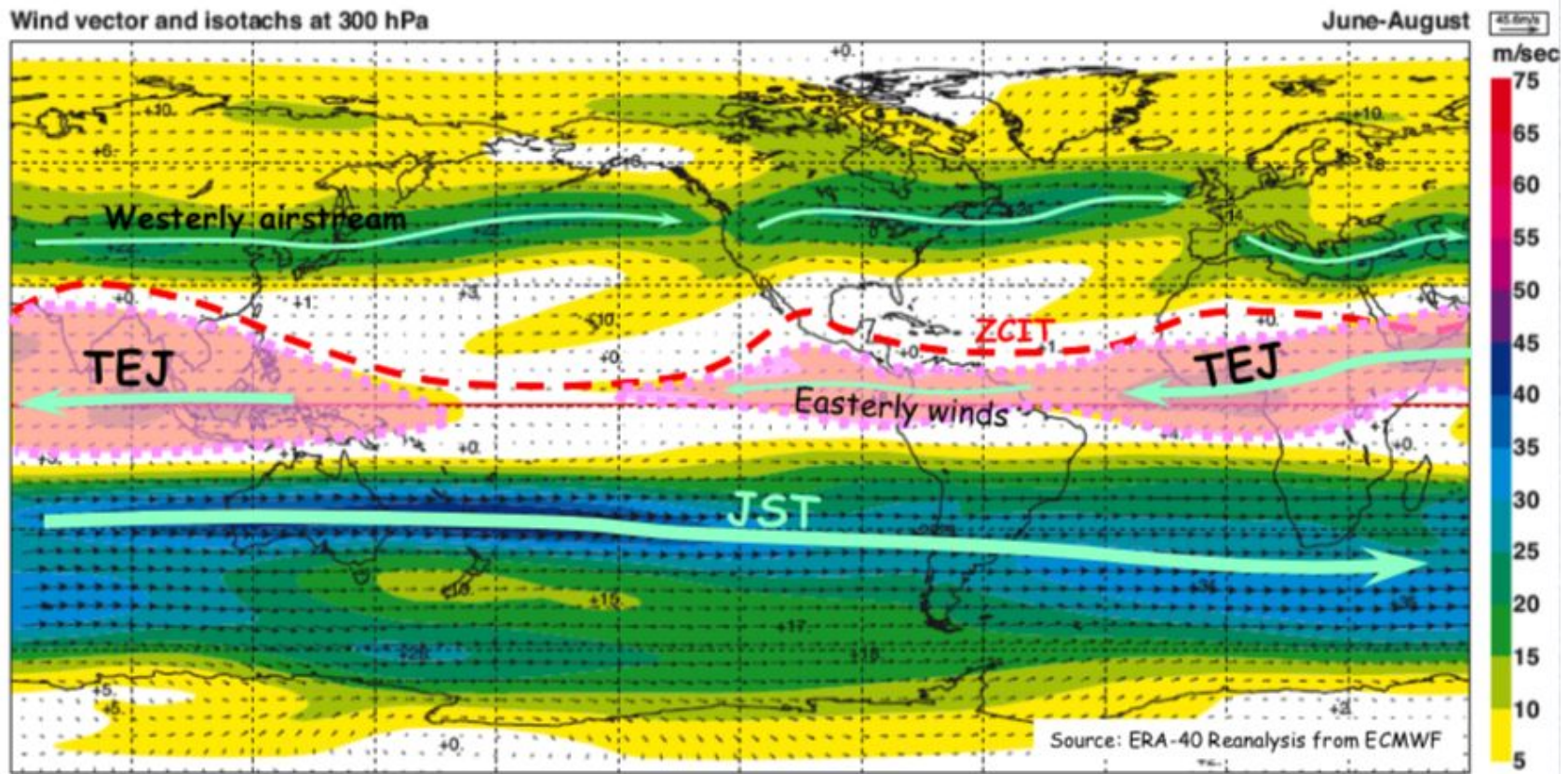
Q: What can you see?



- Fast wave circulation in mid latitudes mainly in Northern hemisphere
- SubTropical Jet (STJ) mainly in Northern hemisphere (can reach 300kts in the East of Asia)
- Light easterly winds near 0°.

At upper level from June to August

Q: What can you see?

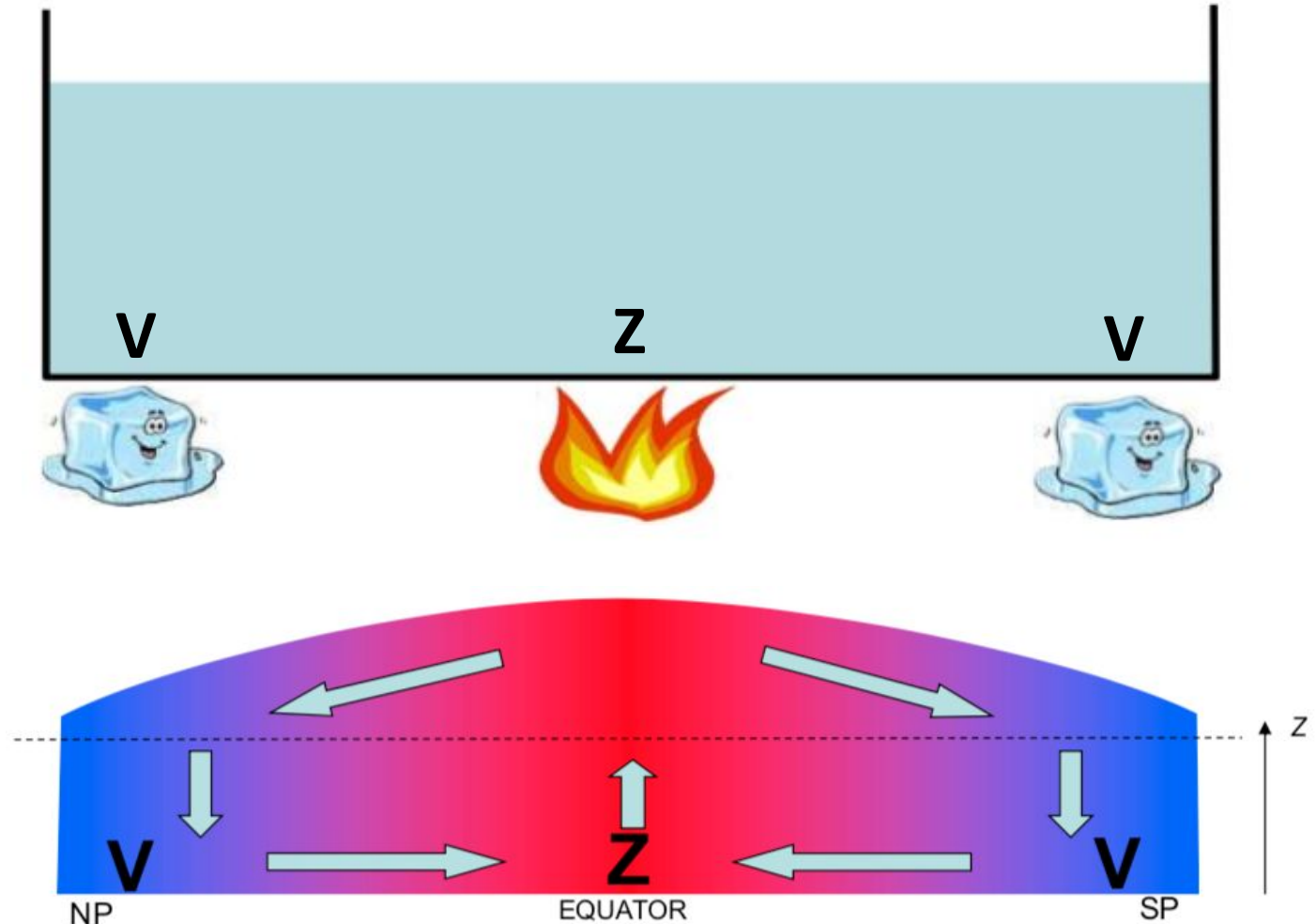


- Weakening of the westerly winds in northern hemisphere
- SubTropical Jet (STJ) mainly in Southern hemisphere
- Tropical Easterly Jet between 20° ~ and 0° , from Asia to Africa

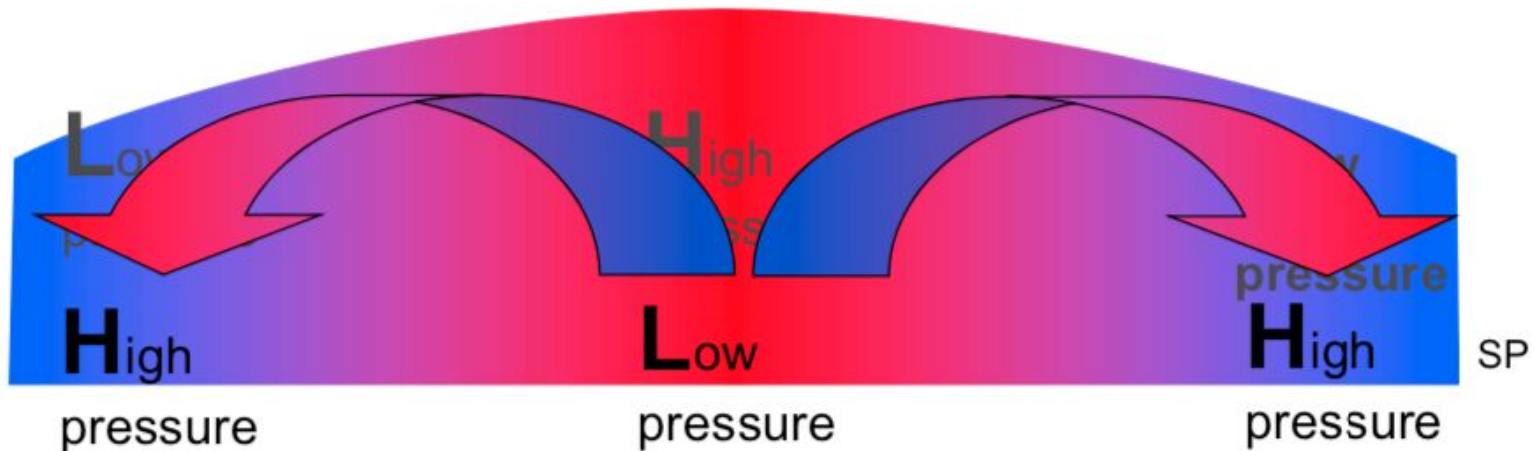
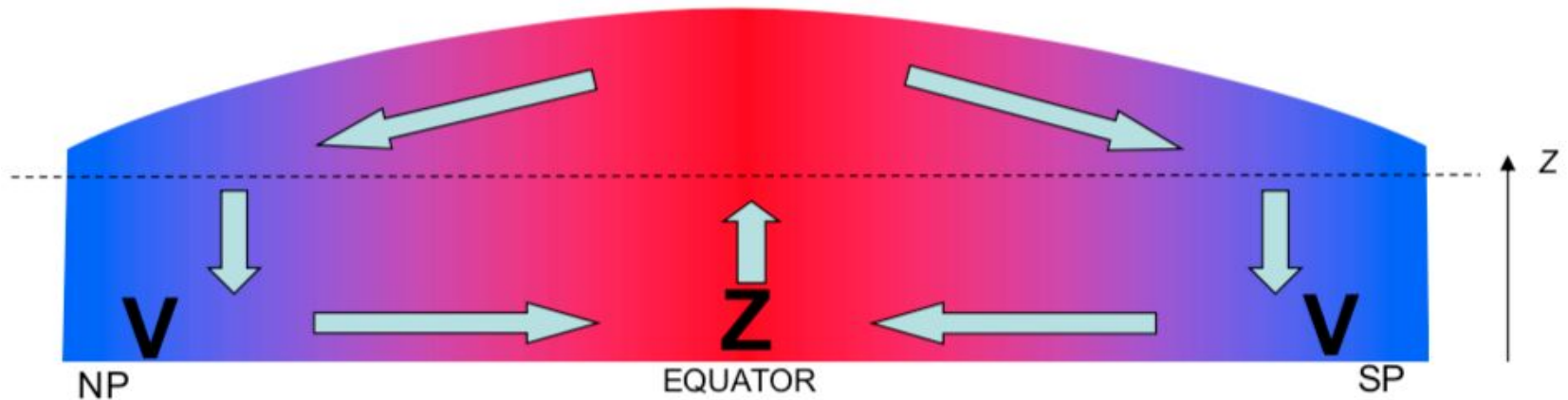
At mean sea level

Q: What can you say about the pressure field observed in Z and V, 2 points located at sea level?

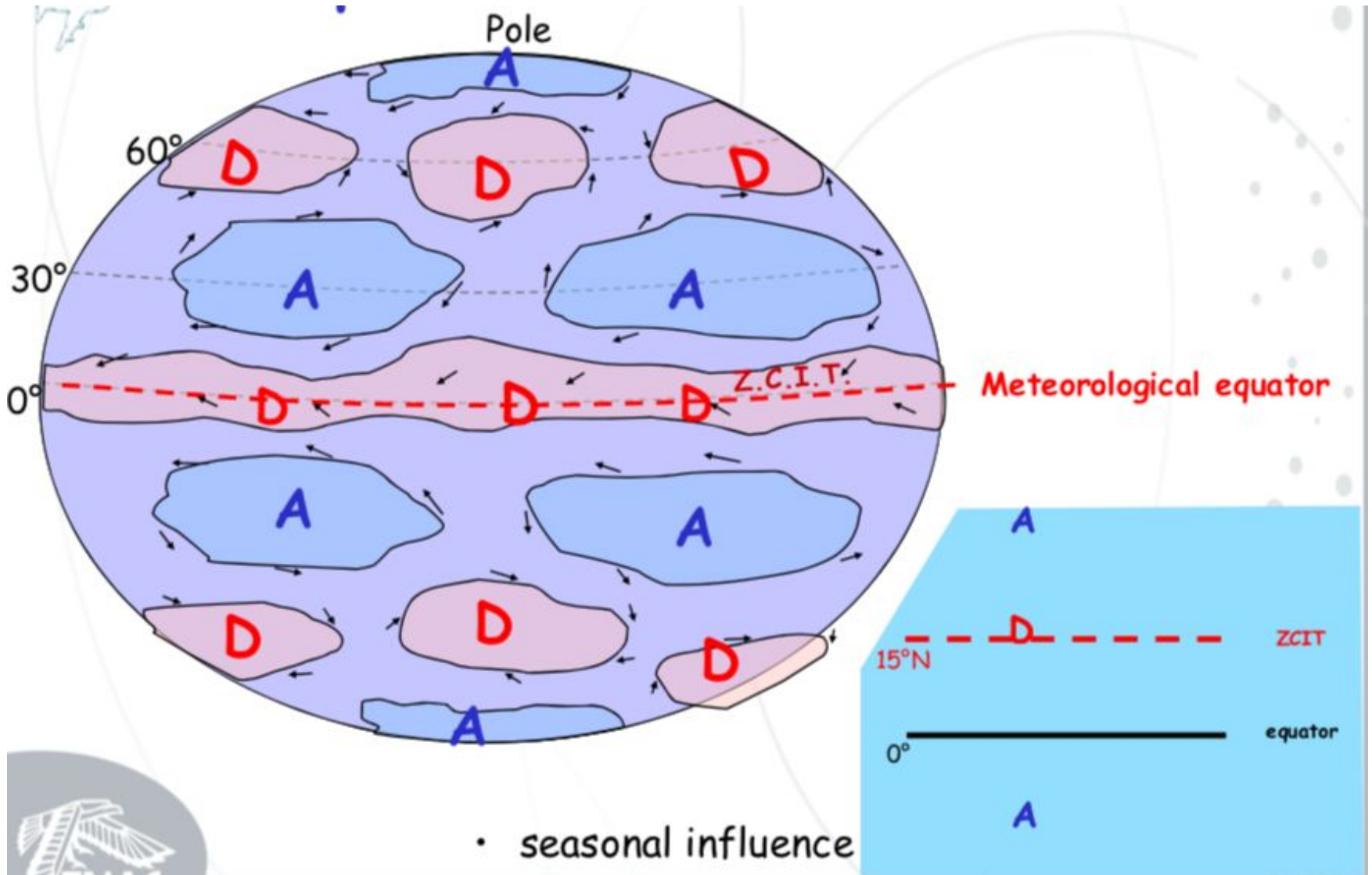
1. $P_z = P_v$
2. $P_z > P_v$
3. $P_z < P_v$



- Cold air having a higher density than hot air → a higher pressure.
- Vertical motion brings mass on polar side, and releases mass on equatorial side.

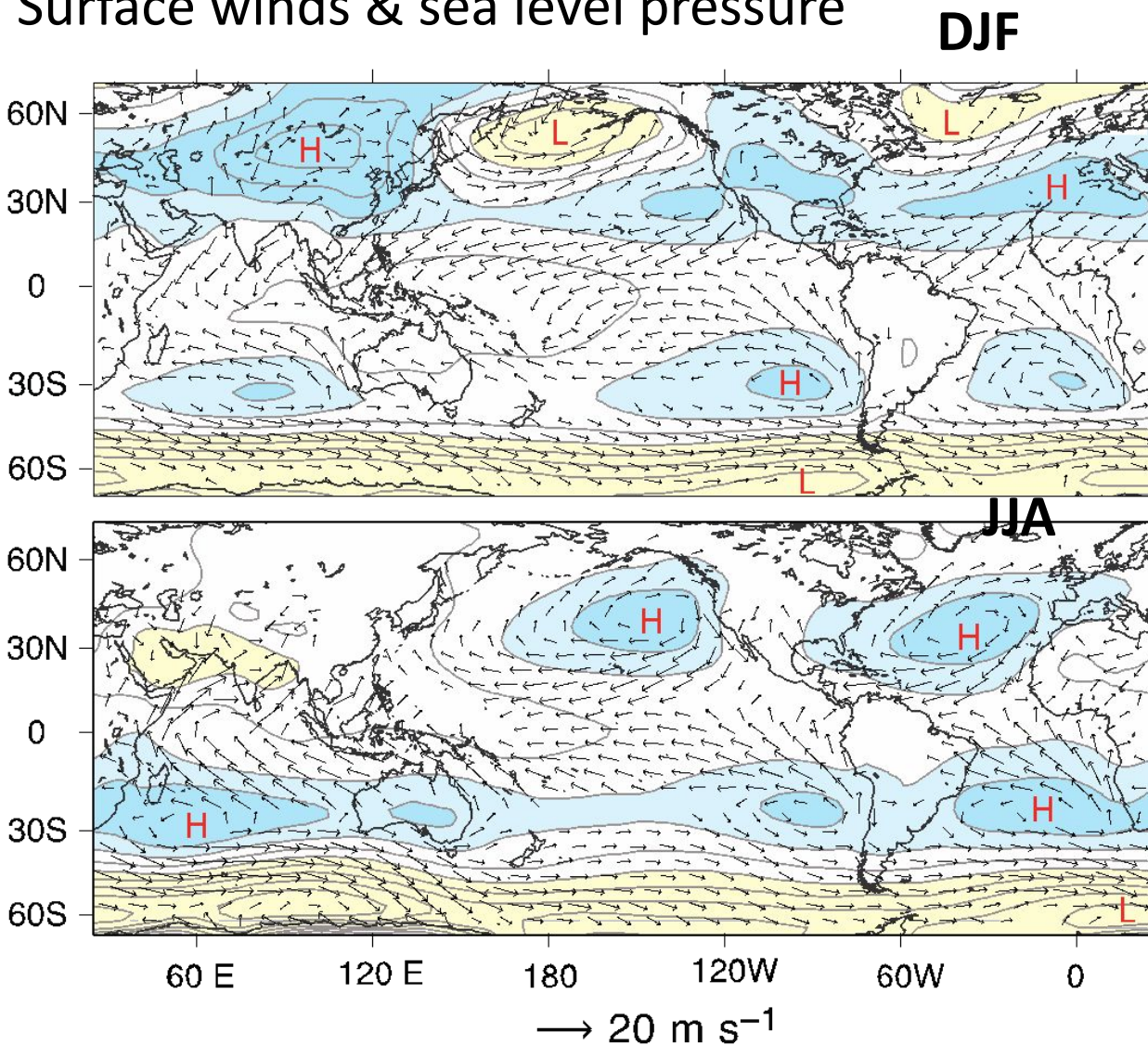


At mean sea level



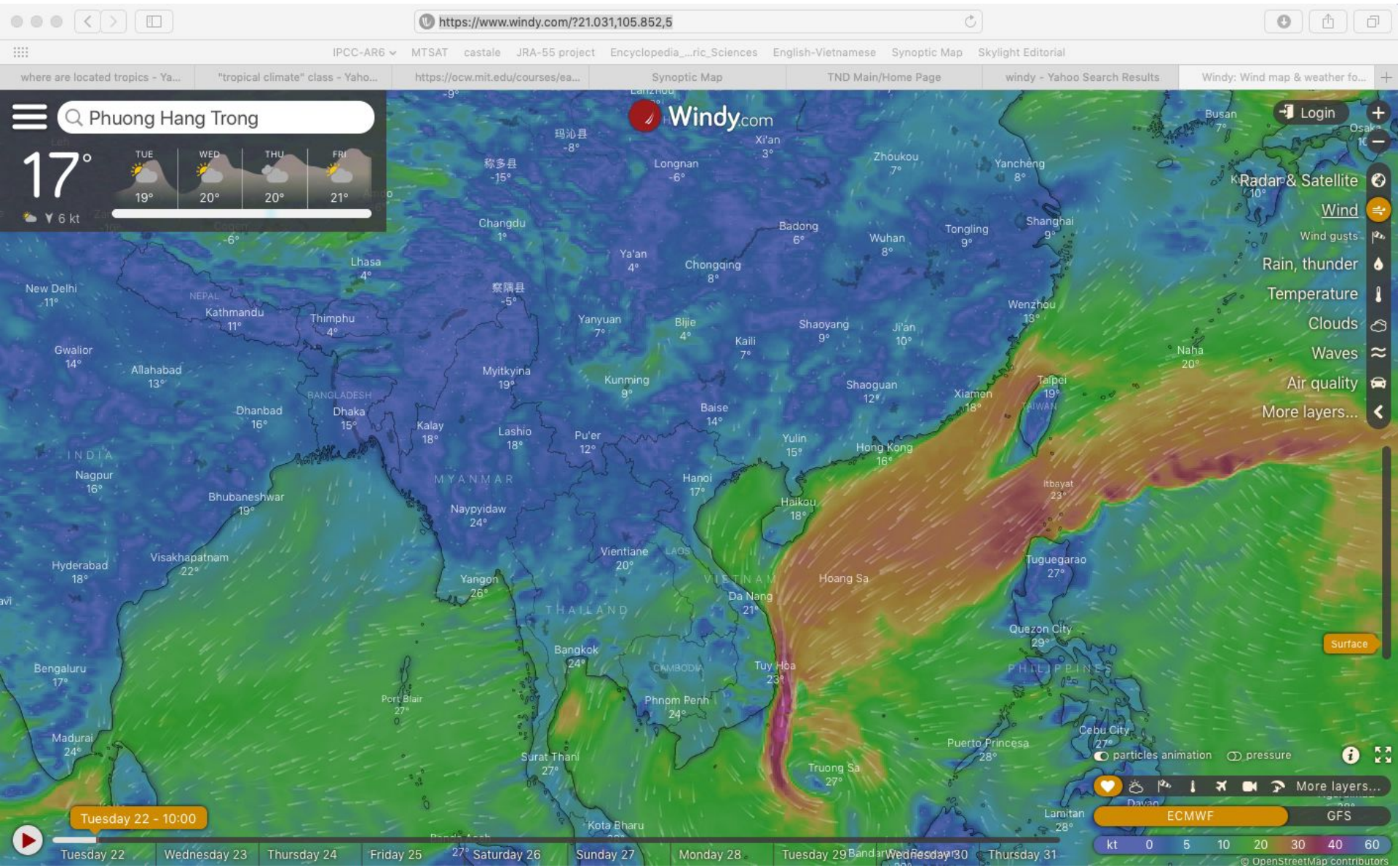
Real world

Surface winds & sea level pressure



- **Atlantic & Pacific Oceans:** surface winds depict well the features shown in the aqua-planet.
- **Southern hemisphere:** ~ aqua-planet
- NH: surface winds strongly influenced by the presence of high latitude continents
- Difference of DJF & JJA

The contour interval for s.l.p is 5hPa. Pressures above 1015hPa are shaded blue, and below 1000hPa are shaded yellow



Dynamics of the atmosphere: The equations

- The dynamics of the atmosphere → in the principles of **conservation of momentum, mass, and energy**
 - The Newton's equations of motion
 - The equation of continuity
 - The thermodynamic energy equation

$$\frac{du}{dt} - \left(f + \frac{u \tan \phi}{a}\right)v + \frac{1}{\rho} \frac{\partial p}{\partial x} + F_x = 0$$

$$\frac{dv}{dt} + \left(f + \frac{u \tan \phi}{a}\right)u + \frac{1}{\rho} \frac{\partial p}{\partial y} + F_y = 0$$

$$p = R\rho T$$

$$\frac{\partial p}{\partial z} + g\rho = 0$$

$$\frac{dT}{dt} + (\gamma - 1)T \nabla \cdot \mathbf{V} = \frac{Q}{c_p}$$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \rho \mathbf{V} = 0$$

$$\frac{\partial \rho_w}{\partial t} + \nabla \cdot \rho_w \mathbf{V} = [\text{sources} - \text{sinks}]$$

- **Independent variables:** space & time coordinates (x,y,z,t)
- **Dependent variables:** velocity, pressure, density, temperature

The hydrostatic equation (see Lecture #2)

$$\frac{du}{dt} - \left(f + \frac{u \tan \phi}{a}\right)v + \frac{1}{\rho} \frac{\partial p}{\partial x} + F_x = 0$$

$$\frac{dv}{dt} + \left(f + \frac{u \tan \phi}{a}\right)u + \frac{1}{\rho} \frac{\partial p}{\partial y} + F_y = 0$$

$$p = R\rho T$$

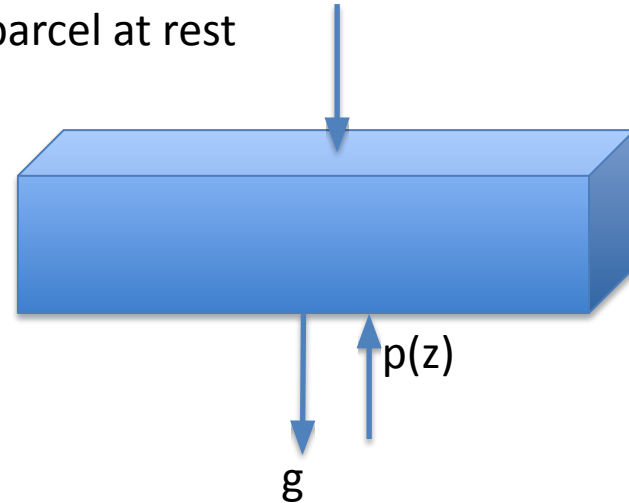
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$$\frac{\partial \rho_w}{\partial t} + \nabla \cdot \rho_w \mathbf{V} = [\text{sources} - \text{sinks}]$$

- In an air parcel at rest



- Force downward: $-[p(z + \Delta z)\Delta x\Delta y + mg]$
- Force upward: $+[p(z)\Delta x\Delta y]$
- Equilibrium:

$$\frac{p(z + \Delta z) - p(z)}{\Delta z} \Delta x\Delta y\Delta z + mg = 0$$

$$\frac{\partial p}{\partial z} + \rho g = 0$$

□ **Hydrostatic balance equation:** balance between the vertical pressure gradient and gravity

Preliminary notions

❖ Eulerian vs. Lagrangian

Eulerian: watch the flow

- Denote the change of a variable with time at **a fixed point** by the Eulerian (or partial) derivative:

$$\frac{\partial p}{\partial t} \quad \text{at a fixed point } (x, y, z)$$

Lagrangian: drift along the flow

- Denote the change of a variable with time **following the flow** by the Lagrangian (or total or material) derivative:

$$\frac{dp}{dt} \quad \text{\& the fluid parcel is fixed}$$

Connection between partial and total derivatives

- For the pressure $p(x(t), y(t), z(t), t)$

$$\frac{dp}{dt} = \frac{\partial p}{\partial t} + \frac{\partial p}{\partial x} \times \frac{\partial x}{\partial t} + \frac{\partial p}{\partial y} \times \frac{\partial y}{\partial t} + \frac{\partial p}{\partial z} \times \frac{\partial z}{\partial t}$$

$$\frac{dp}{dt} = \frac{\partial p}{\partial t} + u \frac{\partial p}{\partial x} + v \frac{\partial p}{\partial y} + w \frac{\partial p}{\partial z}$$

$$\frac{dp}{dt} = \frac{\partial p}{\partial t} + \mathbf{V} \cdot \nabla p \quad (*)$$

Eq (*) is true for all variables

$$\frac{d(\quad)}{dt} = \frac{\partial(\quad)}{\partial t} + \mathbf{V} \cdot \nabla(\quad)$$

Exercise #1

- Given the flow with the speed u in the x -direction

$$u = a \sin(kx - wt)$$

with the constants

a : amplitude

k : wave number

w : frequency

1. Calculate the local and total time derivative of u
2. If $u = 2a \sin(kx - wt)$, how do $\partial u / \partial t$ and du/dt change?

Continuity equation (conservation of mass)

$$\frac{du}{dt} - \left(f + \frac{u \tan \phi}{a}\right)v + \frac{1}{\rho} \frac{\partial p}{\partial x} + F_x = 0$$

$$\frac{dv}{dt} + \left(f + \frac{u \tan \phi}{a}\right)u + \frac{1}{\rho} \frac{\partial p}{\partial y} + F_y = 0$$

$$p = R\rho T$$

$$\frac{\partial p}{\partial z} + g\rho = 0$$

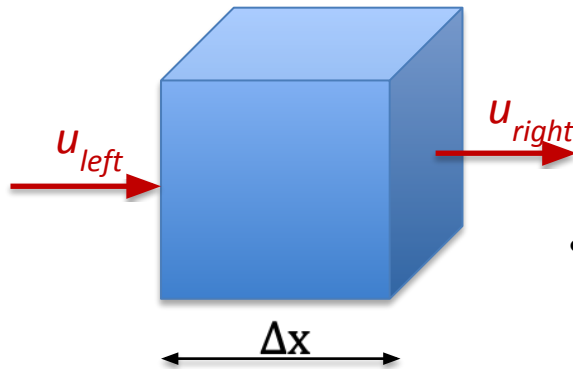
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$$\frac{\partial \rho}{\partial t} + \nabla \cdot \rho \mathbf{V} = 0$$

$$\frac{\partial \rho_w}{\partial t} + \nabla \cdot \rho_w \mathbf{V} = [\text{sources} - \text{sinks}]$$

- **Mass conservation**: the mass of an air parcel remains unchanged with time
- **Continuity equation** is the mathematical expression of **mass conservation**
- **Eulerian & Lagrangian** forms of the continuity equation

Continuity equation: Eulerian formulation



- Consider a cube fixed in space, of dimension $\Delta x = \Delta y = \Delta z$
- The change of mass of the air in the cube

$$\Delta M = \text{mass}_{\text{in}} - \text{mass}_{\text{out}}$$

- For simplicity, consider flow in the x-direction

$$\text{Total mass in the cube: } M = \rho \Delta x \Delta y \Delta z$$

$$\text{Mass change in time } \Delta t: \Delta M = \frac{\partial M}{\partial t} \Delta t = \frac{\partial \rho}{\partial t} \Delta t \cdot V$$

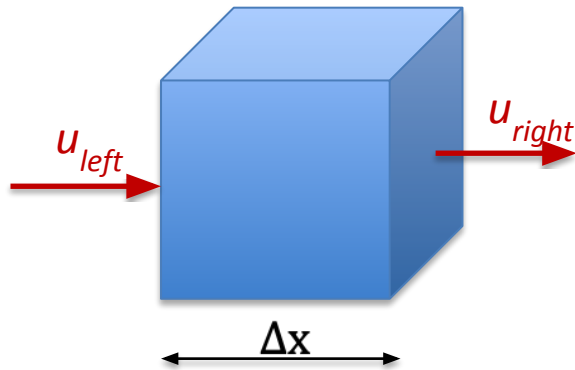
$$\text{Influx from the left: } \rho_{\text{left}} (u_{\text{left}} \Delta t) \Delta y \Delta z = (\rho u)_{\text{left}} \Delta t \Delta y \Delta z$$

$$\text{Outflux on the right: } \rho_{\text{right}} (u_{\text{right}} \Delta t) \Delta y \Delta z = (\rho u)_{\text{right}} \Delta t \Delta y \Delta z$$

$$\text{Net flow into the cube: } F = [(\rho u)_{\text{left}} - (\rho u)_{\text{right}}] \Delta t \Delta y \Delta z$$

$$F = \frac{[(\rho u)_{\text{left}} - (\rho u)_{\text{right}}]}{\Delta x} \Delta t \Delta x \Delta y \Delta z$$

Continuity equation: Eulerian formulation



- $\Delta M = F$

$$\rightarrow \frac{\partial \rho}{\partial t} = \frac{[(\rho u)_{left} - (\rho u)_{right}]}{\Delta x} \approx - \frac{\partial(\rho u)}{\partial x}$$

- Similarly for the two other dimensions y and z

$$\frac{\partial \rho}{\partial t} = - \left(\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} \right)$$

- Using the divergence operator

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{V}) = 0$$

□ The Eulerian form of the continuity equation

Exercise #2

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{V}) = 0$$

Eulerian form

$$\frac{d\rho}{dt} + \rho \nabla \cdot \mathbf{V} = 0$$

Lagrangian form

- Although the two forms look different, but they are equivalent. **Why?**

Incompressibility

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{V}) = 0$$

Eulerian form

$$\frac{d\rho}{dt} + \rho \nabla \cdot \mathbf{V} = 0$$

Lagrangian form

- If the fluid is incompressible (e.g. ocean)
 - the volume of the parcel is unchanged
 - the material density is constant following the flow
 - the continuity equation becomes

$$\nabla \cdot \mathbf{V} = 0$$

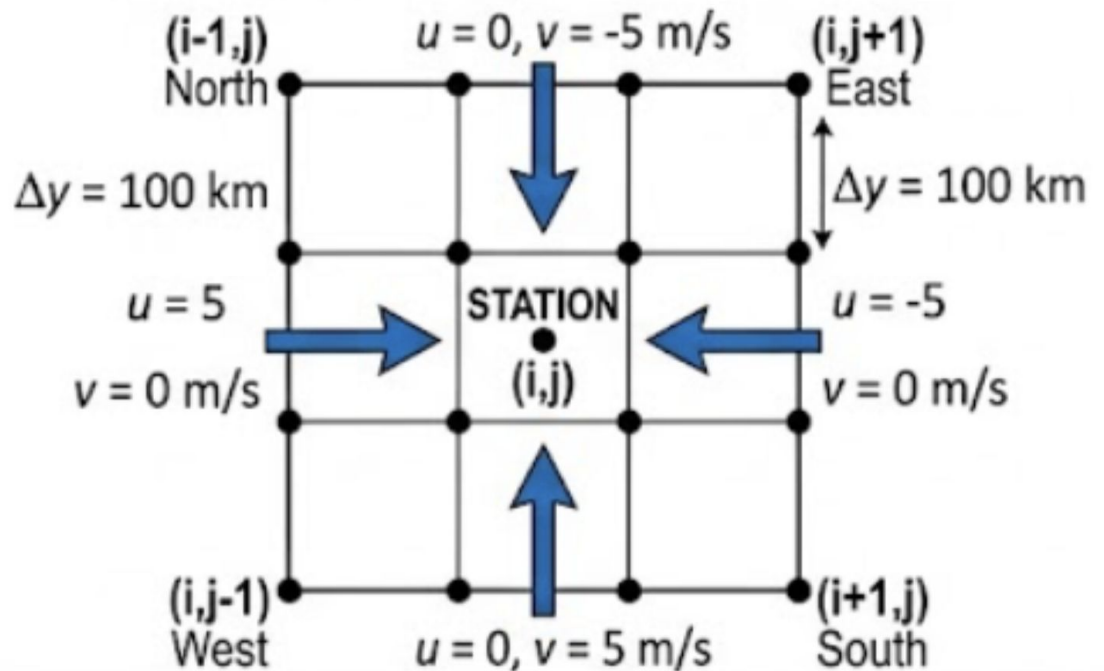
- **Advantages** of assuming incompressibility □ a closed system without having to consider the thermodynamics explicitly.
- **For compressible flow**
 - need another equation, i.e. *the equation of state*, for ρ
 - but this introduces another equation, i.e. *the thermodynamic equation*, for temperature T

Practice #7.1

- Given a 3×3 horizontal weather grid at the top of the planetary boundary layer ($z=1000$ m). The grid spacing (Δx and Δy) is 100 km.
- At the surface ($z=0$ m), the vertical velocity is zero ($w_0=0$).
- Supposing: Incompressible atm.

Write a Python script to calculate the vertical velocity w at the central station at $z=1000$ m.

Will the air be rising or sinking?



Practice #7.2

Open online datasets directly from NOAA

```
url_u = "https://psl.noaa.gov/thredds/dodsC/Datasets/ncp.reanalysis/pressure/uwnd.2026.nc"
```

```
url_v =
```

```
"https://psl.noaa.gov/thredds/dodsC/Datasets/ncp.reanalysis/pressure/vwnd.2026.nc"
```

```
target_time = "2026-03-01T12:00:00"
```

```
target_level = 850
```

Display horizontal wind vector and vertical wind speed (supposing vertical surface wind =0 m/s)